ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** P and R are magnets, Q is magnetic material	Repulsion is seen only between magnets.
2	**(a)** iron is magnetic and copper is not	Only iron is attracted.
3	**(d)** always in the same direction with the same pole of a magnet	Opposite pole forms at the lifting end.
4	**(b)** south-seeking pole	The end that points south is the south-seeking pole.
5	**(a)** iron only	Only iron is magnetic.
6	**(d)** B's north repels A's north	Like poles repel.
7	**(c)** repel	N facing N.
8	**(c)** Earth's magnetism slowly magnetises it	Induction by Earth's field.
9	**(d)** unlike poles attract	Earth's north is magnetic south.
10	**(b)** a magnet or iron object nearby	Local magnetic influence.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The steel table disturbs the needle (1); Move it to a wooden or non-magnetic surface (1).	1+1
12	No (1); Heat weakens the magnet (1).	1+1
13	A pin can be moved through paper by a magnet (1); Non-magnetic sheets do not block force (1).	1+1
14	Nails lose temporary magnetism (1); Soft iron is not a permanent magnet (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Roll bar in filings (1); Maximum cling near the ends (1); Little in the middle (1).	1+1+1
16	Permanent magnet is present (1); A coil moves in its field (1); Produces sound (1).	1+1+1
17	Iron nails are magnetic (1); Brass screws are not magnetic (1); The magnet picks up only the nails (1).	1+1+1
	OR: Gave direction at sea (1); Not dependent on stars (1); Safer voyages (1).	1+1+1
18	Hang a bar magnet freely (1); It settles N-S (1); Repeated turns give same result (1).	1+1+1
	OR: Steel can carry magnetism (1); Nail may be pulled less (1); Magnetic materials change the field (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Same type of clips (1); Same method to approach (1); Count clips lifted by each pole (1); Repeat three times (1); Compare averages (1).	1+1+1+1+1
	OR: Roll magnet in filings (1); Filings gather at the ends (1); Fewer in middle (1); Count or compare quantity (1); Poles are strongest (1).	1+1+1+1+1
20	Test each pair for attraction (1); Copper is attracted by none (1); The magnet may repel another bar's end (1); Iron is attracted by the magnet but not repelled (1); All three identified (1).	1+1+1+1+1
	OR: Heat, impact and time (1); Poor storage (1); Keepers at the poles (1); Pair with unlike poles side by side (1); Cool dry box (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** the pin acts as a temporary magnet	Induced magnetism.
2	**(c)** flat on the palm, away from iron objects	Keep it level and away from magnets.
3	**(c)** the first nail became temporarily magnetised	Induced magnetism.
4	**(a)** stroking in one direction repeatedly	Consistent direction aligns.
5	**(b)** the pin moves toward a magnet held at a distance	It acts across a gap.
6	**(d)** were temporary magnets	Iron loses magnetism easily.
7	**(d)** pull decreases rapidly	Pull weakens with distance.
8	**(c)** repulsion between like poles	Like poles repel.
9	**(d)** it loses magnetism easily	Steel retains.
10	**(c)** breaking a magnet always yields two-pole magnets	Poles come in pairs.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	To check each other's readings (1); Iron in the ship may disturb one (1).	1+1
12	Nails are made of iron, which is magnetic (1); Wood is not a magnetic material (1).	1+1
13	See if it keeps attracting pins after removing the magnet (1); A permanent magnet will (1).	1+1
14	Doctors use a magnet (1); It pulls out iron fragments (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Needle is deflected (1); Aligns along magnet's field (1); Returns when magnet is removed (1).	1+1+1
16	Bring a magnet (1); Only iron sticks (1); Other two do not (1).	1+1+1
17	They may swallow small magnets (1); Magnets can pinch skin (1); They harm gadgets (1).	1+1+1
	OR: Magnets attract steel (1); Hold papers by force (1); No glue needed (1).	1+1+1
18	To prevent rust (1); To protect from damage (1); To give better look (1).	1+1+1
	OR: Magnet pulls pin up (1); Gravity pulls down (1); Pin rises if magnetic force is larger (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	The magnet makes the first clip a temporary magnet (1); That clip makes the next clip a temporary magnet (1); Chain forms (1); On removal, clips lose magnetism (1); They fall (1).	1+1+1+1+1
	OR: Weakening of magnet (1); Wrongly placed poles (1); Dirt reduces grip (1); Check polarity and replace (1); Store properly (1).	1+1+1+1+1
20	Home: fridge door (1); Home: speakers (1); Hospital: scanning devices (1); Industry: separating scrap (1); Transport: motors (1).	1+1+1+1+1
	OR: Compass points north (1); Freely hung magnet aligns (1); Iron objects acquire magnetism (1); Animals navigate by it (1); Evidence is consistent (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** turn away	Like poles repel.
2	**(c)** keeping it with a keeper	Keepers protect.
3	**(c)** away from iron and magnets	Avoid disturbance.
4	**(d)** steel clip, iron nail	Only magnetic items are held.
5	**(a)** suspend it and see which end points north	Free suspension.
6	**(b)** a loudspeaker	It contains a magnet.
7	**(c)** 3	Each piece is a magnet with two poles.
8	**(c)** a magnet	Only a magnet has both poles.
9	**(d)** it turns to align with the field	It aligns with the combined field.
10	**(b)** increasing the distance	Force reduces with distance.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Prevents unwanted sticking (1); Prevents loss of strength (1).	1+1
12	Cutting a magnet gives two magnets (1); Each has N and S poles (1).	1+1
13	They are non-magnetic (1); They do not disturb the magnets (1).	1+1
14	It attracts all pins at once (1); It saves time (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	It weakens (1); Alignment is disturbed (1); Handle gently (1).	1+1+1
16	Knife holders (1); Spice tins (1); Fridge doors (1).	1+1+1
17	Heat a weak magnet gently (1); Test with pins before and after (1); It picks fewer pins (1).	1+1+1
	OR: Bring near pins (1); Pins stick if magnetised (1); Check poles with compass (1).	1+1+1
18	Needle deflects strongly (1); May reverse (1); Needle may lose accuracy (1).	1+1+1
	OR: Steel is attracted by the magnet (1); The magnet turns towards the table (1); It gives a wrong direction (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Magnet does not attract gold (1); Gold is non-magnetic (1); Magnet cannot separate it (1); Use water to wash away sand (panning) (1); Gold is heavier and settles (1).	1+1+1+1+1
	OR: Place needle near one end (1); Needle's north end is attracted (1); That end is S pole (1); Check the other end similarly (1); Record (1).	1+1+1+1+1
20	Needle is a tiny magnet (1); Earth is like a giant magnet (1); Needle aligns with Earth's field (1); Gives direction (1); Used in ships and planes (1).	1+1+1+1+1
	OR: Temporary: soft iron near magnet (1); Permanent: steel bar magnet (1); Natural: lodestone (1); Differences in retention (1); Uses differ (1).	1+1+1+1+1

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